

# Six group point reactor kinetics simulation

Technical report

## Summary

This project implements a seven-variable point reactor kinetics model in Python to study normalized fission power following prescribed reactivity changes. Six delayed-neutron precursor groups couple to the neutron population. The calculations show a rapid initial adjustment followed by slower precursor-dependent evolution. Input history matters: a step and a ramp reaching the same reactivity produce different power histories.

For the illustrative baseline, a +0.10-dollar step produces a normalized power of 2.341293 at 60 seconds; a ten-second ramp to the same target produces 2.217767. A -0.10-dollar step reduces power to 0.542138. Shortening prompt neutron generation time from 100 to 10 microseconds accelerates the initial adjustment but changes the 60-second positive-step result by only about 0.18 percent.

The implementation passes equilibrium checks, comparisons with matrix-exponential solutions, and selected tolerance-refinement checks. These establish numerical consistency for the tested cases. No experimental validation or reactor-specific calibration has been performed.

## Purpose and scope

The research question is how prescribed reactivity and delayed-neutron dynamics shape the time response of fission power. The project is an educational computational study suitable for demonstrating equation implementation, parameter documentation, verification, and interpretation.

The model represents the core with one neutron amplitude rather than a spatial distribution. It does not calculate transport, geometry, enrichment, criticality searches, fuel temperature, coolant behavior, electrical generation, or decay heat. No result is a prediction for a named reactor or a plant safety assessment.

## Main findings

- Positive reactivity held constant causes continuing growth in this fixed-parameter model.
- Early-time responses agree closely with the prompt-jump approximation in the two small positive steps examined.
- Fast precursor groups follow changing production more closely than slow groups.
- A gradual insertion gives less accumulated growth than an immediate step to the same target.
- Prompt generation time strongly affects the early adjustment in these cases, while the longer response remains strongly influenced by delayed neutrons.

## Governing equations

The seven states are the normalized neutron amplitude  $n$  and six precursor amplitudes  $C_i$ . Both are divided by the initial neutron population  $n_0$ . Under the assumed fixed flux shape and fission properties, normalized fission power  $P/P_0$  equals  $n$ .

$$\begin{aligned}dn/dt &= ((\rho - \beta)/\Lambda)n + \sum_i \lambda_i C_i \\dC_i/dt &= (\beta_i/\Lambda)n - \lambda_i C_i \quad (i = 1, 2, \dots, 6) \\ \beta &= \sum_i \beta_i ; \quad \rho = (k_{eff} - 1)/k_{eff}\end{aligned}$$

These are the conventional point-kinetics balance equations [1]. The net prompt term and delayed-neutron source determine the neutron rate of change. Precursor production is proportional to  $n$ ; precursor loss is proportional to the group population.

| Symbol      | Meaning                                       | Units         |
|-------------|-----------------------------------------------|---------------|
| $n$         | Neutron amplitude divided by $n_0$            | Dimensionless |
| $C_i$       | Precursor amplitude divided by $n_0$          | Dimensionless |
| $\rho$      | Reactivity                                    | Dimensionless |
| $\beta_i$   | Effective delayed fraction used for group $i$ | Dimensionless |
| $\lambda_i$ | Group decay constant                          | $s^{-1}$      |
| $\Lambda$   | Prompt neutron generation time                | $s$           |

One dollar of reactivity means  $\rho = \beta$ . Here +0.10 dollars means  $\rho = 0.0006502$ , or 65.02 pcm, where one pcm equals  $10^{-5}$  in dimensionless reactivity. It does not mean a ten-percent change in power. Prompt generation time is not an interchangeable name for prompt neutron lifetime.

## Initial equilibrium

Every experiment starts from a critical equilibrium immediately before the input change. Set  $n(0) = 1$  and  $C_i(0) = \beta_i/(\Lambda\lambda_i)$ . The precursor equation then gives zero rate of change. At  $\rho = 0$ , the neutron equation also gives zero because the delayed source  $\beta/\Lambda$  exactly balances the net prompt term  $-\beta/\Lambda$ .

Zero initial precursor populations would represent a different physical initial condition and would create an unwanted transient. The generation-time study recalculates equilibrium precursor populations separately for each  $\Lambda$ .

## Parameters and numerical method

The group entries below are taken from the thermal U-235 data in Table 1 of Ragheb [2]. Their use as effective kinetics fractions is an educational approximation: no reactor-specific neutron importance weighting is calculated.

| Group | $\beta_i$ | $\lambda_i$ in $s^{-1}$ | $1/\lambda_i$ in seconds |
|-------|-----------|-------------------------|--------------------------|
| 1     | 0.000215  | 0.0124                  | 80.65                    |
| 2     | 0.001424  | 0.0305                  | 32.79                    |
| 3     | 0.001274  | 0.111                   | 9.01                     |
| 4     | 0.002568  | 0.301                   | 3.32                     |
| 5     | 0.000748  | 1.14                    | 0.88                     |
| 6     | 0.000273  | 3.01                    | 0.33                     |

The sum  $\beta = 0.006502$  retains the rounded table entries, rather than substituting the table's rounded total 0.0065. The fourth decay constant is  $0.301 s^{-1}$ ; entries from different parameter sets are not mixed. Baseline  $\Lambda = 10^{-4} s$  is an illustrative assumption. The sensitivity case uses  $10^{-5} s$ .

SciPy `solve_ivp` integrates the equations using Radau, an implicit method recommended for stiff systems [3]. Prompt and precursor dynamics have widely separated timescales. Baseline relative tolerance is  $10^{-8}$  and absolute tolerance is  $10^{-10}$ . Selected checks tighten these to  $10^{-10}$  and  $10^{-12}$ .

The plotted times are requested output samples, not fixed internal solver steps. Dense output evaluates the solution at those times. Steps occur at  $t = 0$ ; the state remains continuous. The ten-second ramp increases linearly from zero and then holds. The solver restarts at its endpoint to handle the change in input slope explicitly.

## Verification evidence

- Equilibrium preservation checks the physical initial condition and the constant-state solution.
- For both signs of the baseline step, all seven states are compared with  $\exp(A t)y_0$  at selected times. The matrix is assembled separately from the derivative function.
- Baseline step and ramp results are compared with tighter solver tolerances using a relative agreement criterion of  $2 \times 10^{-7}$ .
- The ramp's constant-input segment from 10 to 60 seconds is checked against a matrix-exponential continuation from its endpoint state.
- The shorter-generation-time case passes equilibrium and matrix-exponential checks.

The executable checks pass. Matrix agreement tests the integration and implementation; it is not independent evidence for the underlying physical assumptions. The prompt-jump comparison below is a separate approximation check.

# Reactivity magnitude and early response

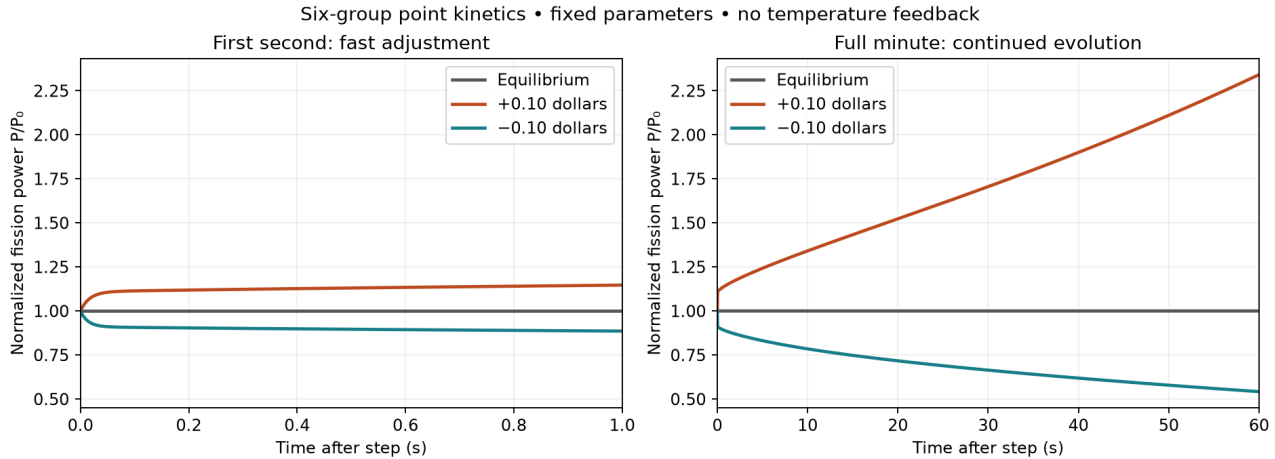


Figure 1. Baseline equilibrium and signed steps. The left panel displays the first second; the right panel displays a full minute. Power remains normalized to each experiment's starting value.

A +0.20-dollar step was also evaluated with the same model. Doubling reactivity does not double power at a specified later time because the population-dependent growth accumulates over time.

| Time in seconds | +0.10 dollars | +0.20 dollars |
|-----------------|---------------|---------------|
| 0               | 1.000000      | 1.000000      |
| 10              | 1.341112      | 1.933656      |
| 30              | 1.705738      | 3.519822      |
| 60              | 2.341293      | 8.030966      |

For a step from critical equilibrium, the frozen-precursor prompt-jump estimate is  $\beta/(\beta-\rho)$ . It predicts 1.111111 for +0.10 dollars and 1.250000 for +0.20 dollars. At 0.10 seconds, the numerical values are 1.113976 and 1.256221, respectively: approximately 0.258% and 0.498% above the estimates.

This comparison concerns an early response window after the fast adjustment, before substantial precursor evolution. The approximation does not predict a plateau or the power at 60 seconds. The full solution is continuous, and its precursors are never actually frozen. Agreement at one selected time is not a uniform error bound over the transient.

## Step and ramp comparison

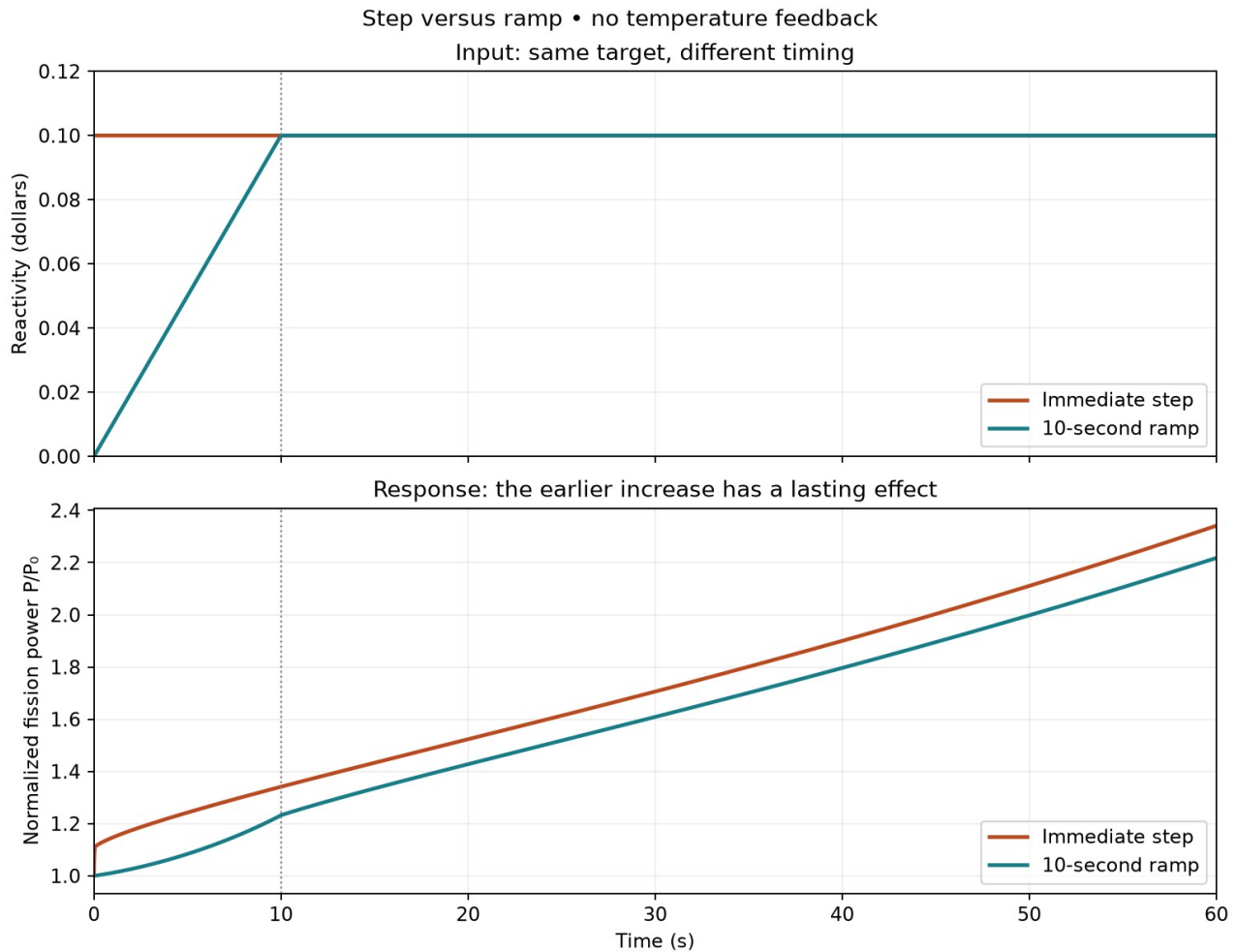


Figure 2. The ramp reaches +0.10 dollars at ten seconds. The input curves coincide afterward, but the power curves do not.

| Time in seconds | Immediate step $P/P_0$ | Ten second ramp $P/P_0$ |
|-----------------|------------------------|-------------------------|
| 10              | 1.341112               | 1.231656                |
| 30              | 1.705738               | 1.608566                |
| 60              | 2.341293               | 2.217767                |

The step supplies higher reactivity throughout the interval before ten seconds. Neutrons and precursors therefore accumulate differently. Equal reactivity at and after ten seconds does not erase differences in the state. This supports the original prediction that the immediate step produces higher power at ten seconds.

The ramp is a prescribed reactivity history. Calling it a physical control-rod withdrawal would require a relationship between rod position and reactivity that this model does not contain.

## Precursor behavior during rising and falling power

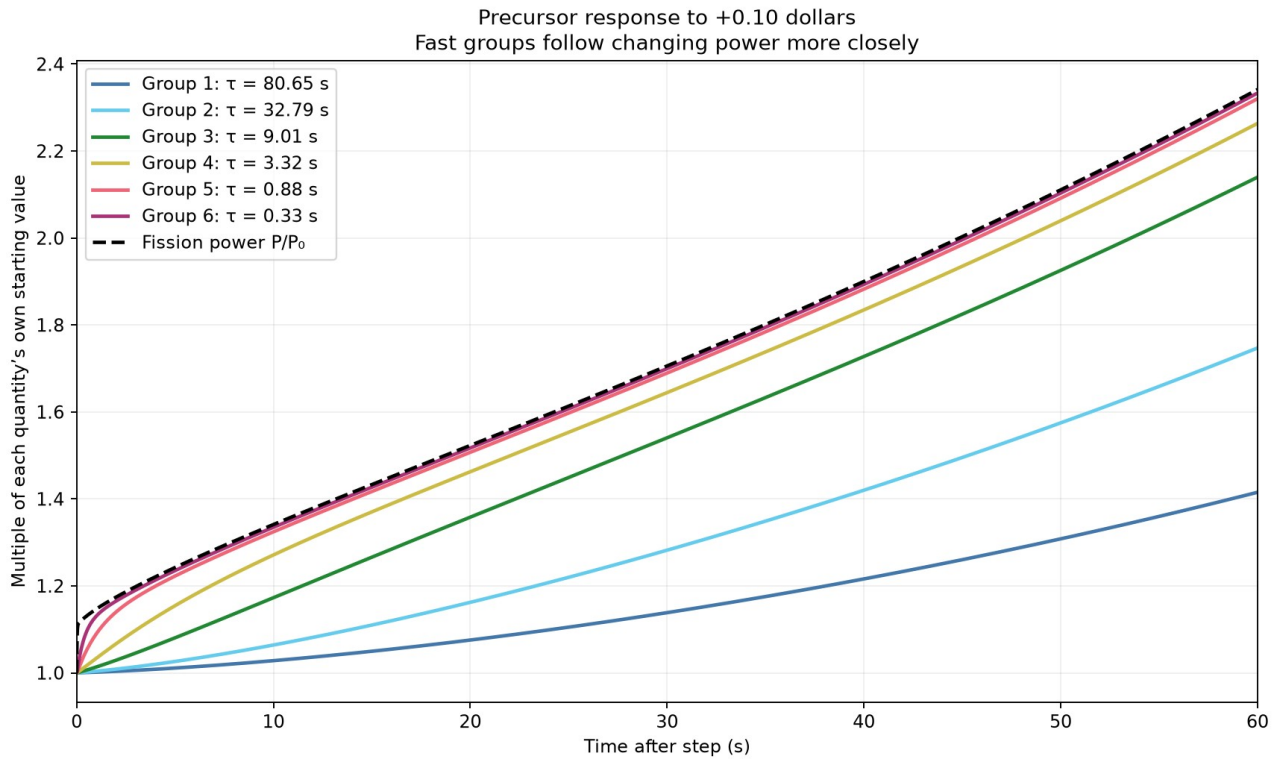


Figure 3. Each precursor curve is divided by its own initial value. Faster groups track increasing production more closely.

| Quantity                    | +0.10 dollars at 60 s | -0.10 dollars at 60 s |
|-----------------------------|-----------------------|-----------------------|
| Fission power $P/P_0$       | 2.341293              | 0.542138              |
| Group 1 $C_1/C_1$ initially | 1.415224              | 0.821307              |
| Group 2                     | 1.747055              | 0.692715              |
| Group 3                     | 2.139783              | 0.576900              |
| Group 4                     | 2.263429              | 0.554126              |
| Group 5                     | 2.320266              | 0.545236              |
| Group 6                     | 2.333288              | 0.543306              |

Group 6 has a short characteristic time and closely tracks the changing production rate. Group 1 adjusts more slowly. Curve height compares proportional changes, not absolute group populations or contributions to the neutron source.

# Negative reactivity and retained precursor populations

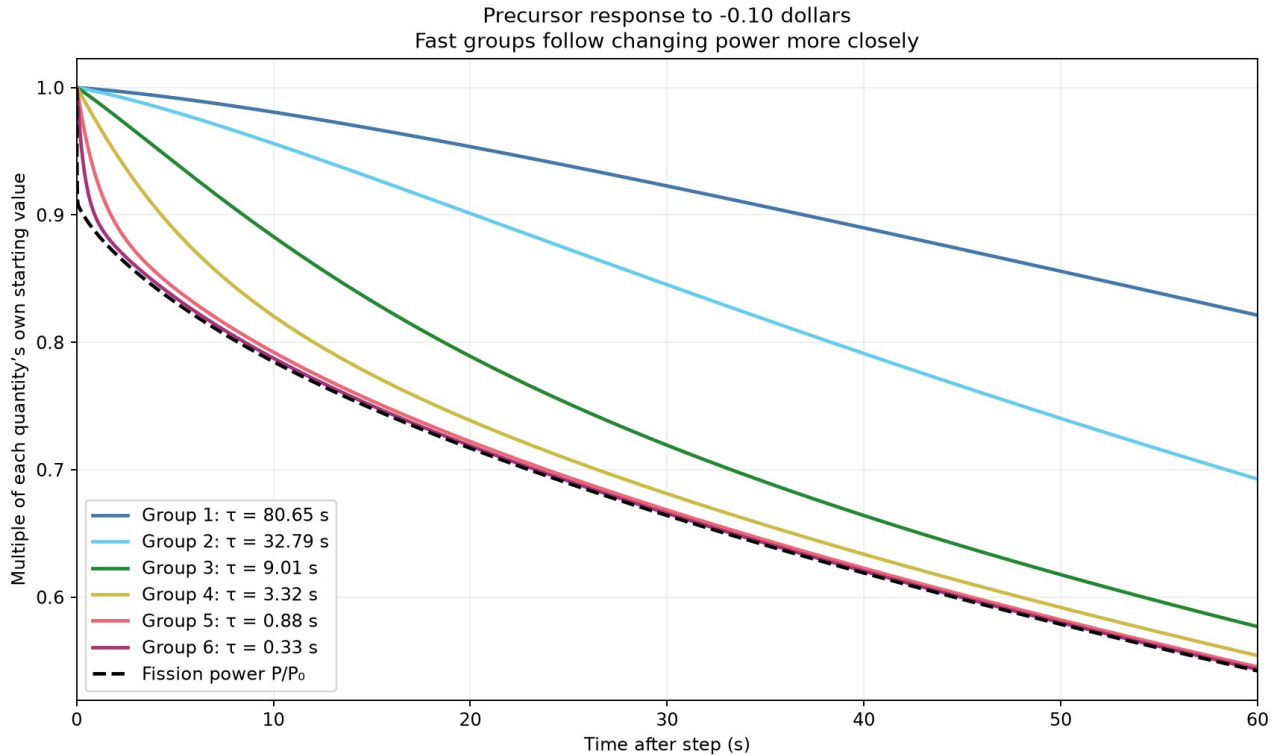


Figure 4. Following the negative step, fast groups decline close to power while slow groups retain more of their initial populations.

The increased net neutron loss initially reduces power. Existing precursors continue supplying delayed neutrons, while reduced fission lowers new precursor production. Slow groups retain the influence of preceding steady operation for longer. A large normalized group-1 population does not establish that group 1 supplies the most neutrons; that would require comparing  $\lambda_i C_i$ .

This is a small negative step from an idealized equilibrium, not a calibrated shutdown sequence. The power shown is fission power. Radioactive decay heat, stored thermal energy, and cooling-system response are absent, so the calculation cannot determine total heat removal requirements.

## Meaning of the group timescales

For an isolated group with a sudden change to constant production  $S$ ,  $dC/dt = S - \lambda C$  has solution  $C(t) = S/\lambda + [C(0) - S/\lambda] \exp(-\lambda t)$ . The characteristic time is  $\tau = 1/\lambda$ . One  $\tau$  closes about 63.2% of the gap; three close about 95.0%; five close about 99.3%.

In these coupled reactor experiments, production continues changing. The timescales describe responsiveness rather than times at which the groups finish adjusting to a fixed final population.

# Prompt generation time sensitivity

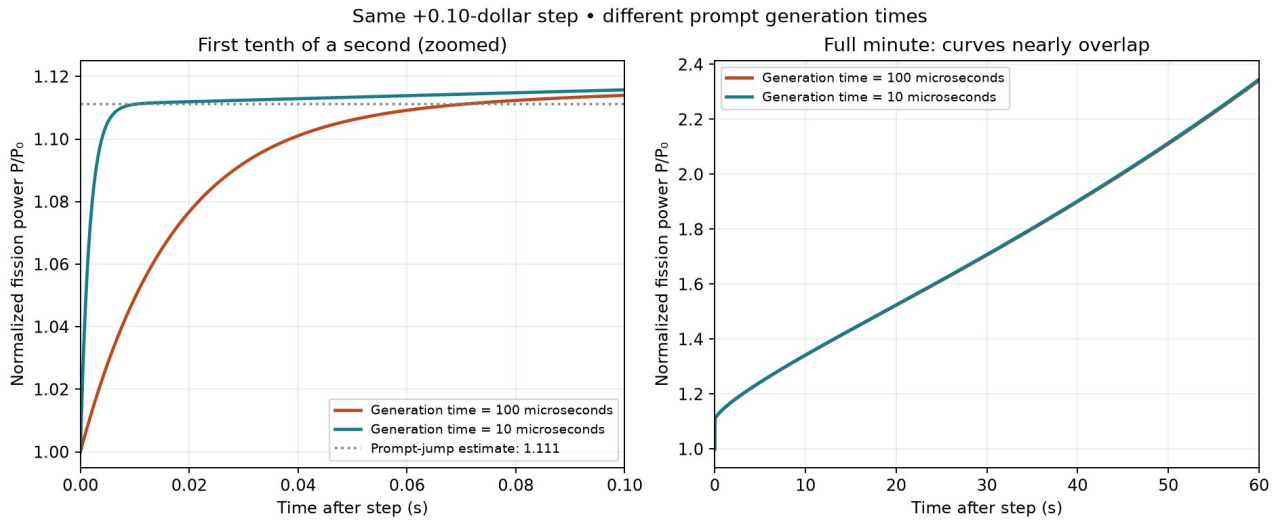


Figure 5. A shorter prompt generation time accelerates the early adjustment. The left panel zooms both time and power; its vertical scale differs from the right panel.

| Time in seconds | $\Lambda = 100$ microseconds | $\Lambda = 10$ microseconds |
|-----------------|------------------------------|-----------------------------|
| 0.01            | 1.049243                     | 1.111122                    |
| 0.10            | 1.113976                     | 1.115733                    |
| 60              | 2.341293                     | 2.345570                    |

The early frozen-precursor relaxation time is approximately  $\Lambda/(\beta-p)$ , about 0.0171 s and 0.00171 s for these cases. The prompt-jump amplitude estimate remains the same because  $p/\beta$  is unchanged. The shorter case approaches it sooner.

At 60 seconds the difference is about 0.18%. Shortening  $\Lambda$  by a factor of ten therefore does not accelerate the entire response by ten. This conclusion applies to the small delayed-supercritical insertion studied, not all possible reactivity regimes.

## Conclusions and limitations

The model reproduces the expected qualitative behavior and passes numerical benchmarks for the tested histories. Its central finding is that prompt and delayed timescales jointly shape power transients, while the state retains the effects of the preceding reactivity history.

Limitations include fixed coefficients, an assumed fixed spatial shape, approximate effective delayed fractions, no thermal feedback, no burnup or poisoning, no external source, and no decay heat. Unbounded growth under sustained positive reactivity is a feature of these assumptions, not an operational forecast. Experimental validation and reactor-specific calibration remain outside the completed scope.

## Reproducing the project

The project uses Python 3.14, NumPy 2.5.3, SciPy 1.18.1, and Matplotlib 3.11.2. The delivered requirements.txt pins the three numerical and plotting packages. From the project directory, create an isolated environment and run:

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements.txt
python kinetics.py
```

The script runs its checks before generating five plots. A failed assertion or solver failure prevents a clean successful run. The figures are first\_results.png, step\_vs\_ramp.png, precursor\_response.png, negative\_precursor\_response.png, and generation\_time\_comparison.png.

The +0.20-dollar and prompt-jump tables can be reproduced using the existing public function:

```
import numpy as np
from kinetics import BETA, simulate
times = np.array([0., .01, .05, .1, 10., 30., 60.])
for dollars in (.1, .2):
    power = simulate(dollars * BETA, times)[0]
    estimate = 1 / (1 - dollars)
    error = 100 * abs(power[3] / estimate - 1)
    print(dollars, power, estimate, error)
```

## References

[1] Argonne National Laboratory. SAM theory manual, Chapter 6, Reactor Power Models. Governing point-kinetics formulation.

<https://publications.anl.gov/anlpubs/2021/05/166771.pdf>

[2] M. Ragheb. Point Reactor Kinetics, March 30, 2014. Table 1, PDF page 10, thermal U-235 group entries.

<https://www.mragheb.com/NPRE%20402%20ME%20405%20Nuclear%20Power%20Engineering/Point%20Reactor%20Kinetics.pdf>

[3] SciPy documentation. scipy.integrate.solve\_ivp. Solver methods, tolerances, and dense output.

[https://docs.scipy.org/doc/scipy/reference/generated/scipy.integrate.solve\\_ivp.html](https://docs.scipy.org/doc/scipy/reference/generated/scipy.integrate.solve_ivp.html)

## Project provenance

This is an AI-assisted educational project developed through guided discussion, code generation, execution, and interpretation. The report describes computational results and does not claim professional reactor-engineering experience or independent experimental validation. Any portfolio description should accurately reflect the learner's own understanding and contributions.